PAPER

Locating and Mitigating Factual Hallucination Circuits in Small Language Models via Activation Patching

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SUMMARY Small language models (SLMs) are increasingly deployed in resource-constrained systems, yet they remain prone to factual hallucination, particularly when a misleading context conflicts with parametric knowledge (a context–memory conflict). We present a systematic activation-patching study of this conflict in three SLMs: Gemma-2-2B (2,000 facts), Llama-3.2-1B (600 facts), and Qwen2.5-1.5B (600 facts), each filtered to facts the model demonstrably knows. Restoration effects concentrate in the final quarter of the network across all three architectures (pairwise Spearman rank correlation of layer profiles 0.81–1.00), with a small set of late-layer attention heads dominating. Bidirectional patching reveals that influential heads are superposed—each both conveys the misleading context and carries the memorized answer—and separates dual-role workhorses from conflict-specialist heads whose neutral-control effect is minimal. In Gemma-2-2B, knocking out the single specialist head L22.H4 reduces misleading-context distractor-following by 35% relative while increasing factual accuracy by 5.3 percentage points (McNemar $p < 10^{-23}$), with MMLU retention 0.998 and a 5% perplexity cost; single-head knockouts from the patching ranking reduce distractor-following by 30–50% in the other two models as well. In contrast, norm-based head selection *increases* distractor-following. We report quantified dose–response trade-offs between distractor-following reduction and capability, using a protocol that guards against circular analysis via relation-level splits.

key words: *language models, hallucination, activation patching, attention heads*

1. Introduction

Language models of 1–2.6B parameters now power on-device assistants, retrieval-augmented pipelines, and document-processing systems where latency and memory budgets preclude larger models. In such deployments the model must frequently integrate an external context with its parametric knowledge. When the context asserts a false fact—whether through a retrieval error, an adversarial document, or a mistaken user—the model faces a *context–memory (CM) conflict* [5], and resolving it incorrectly produces a factual hallucination.

Why a mechanistic account rather than a purely behavioral mitigation? Deployed systems need interventions that are cheap, targeted, and auditable: knowing *which* components carry the conflict lets a system builder trade hallucination against capability with measured confidence, instead of retraining, prompt-tuning blindly, or trusting an opaque detector. The account is more valuable still if the responsible components—and hence the mitigation recipe—transfer across architectures, which is why replication is a first-class concern in this work rather than an afterthought.

A growing body of work studies hallucination mechanisms, including knowledge-deficit failures [2] and probes/directions for truthfulness [3], [4]. Head-level analyses of CM conflict, however, have appeared mainly in multimodal settings [14], and recent work on knowledge-conflict interventions reports that influential attention heads play *superposed* dual roles [13]. What has been missing for sub-2B SLMs is a systematic causal account: which components, located how, trade hallucination against general capability when intervened upon, and whether such findings replicate across architectures.

This paper provides that account with a design that separates localization from validation: heads are selected only on a development split of relations, and all intervention results are reported on held-out relations. Our contributions are:

- **Localization.** A coarse-to-fine activation-patching map of the CM-conflict circuit in Gemma-2-2B (and its replication in two other architectures): restoration effects concentrate in late layers, and the attention/MLP decomposition shows the conflict is attention-dominated, unlike the MLP-heavy picture of classical fact recall.
- **Characterization.** Bidirectional patching quantifies superposition per head and separates dual-role heads from conflict-specialist heads with minimal neutral-control effect.
- **Quantified intervention trade-offs.** Dose–response curves for knockout and mean-ablation, with fairness baselines: patching-guided selection reduces hallucination significantly (McNemar $p < 10^{-6}$ even for a single head), whereas activation-norm selection *increases* it; capability costs are dissociated (MMLU retention ≈ 1.00; perplexity +11–20%).

2. Related Work

Hallucination and knowledge conflict. Surveys taxonomize hallucination in NLG [6] and knowledge conflicts in LLMs [5]. Behavioral studies quantify when context overrides memory [7]; Chuang et al. detect contextual hallucinations from attention maps [8]. Yu et al. trace non-factual hallucination to knowledge deficits in MLPs and selection failures in attention [2]; our corruption regime instead tar-

Manuscript received October 15, 2026.
Manuscript revised October 15, 2026.

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DOI: 10.1587/trans.E0.??.1

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gets *knows-but-overridden* failures. Mechanistic work has also examined conflicts inside parametric memory itself [18] and cautioned that heads appearing to implement factual recall may partly implement generic copying [19]—a confound our neutral-control design quantifies directly.

Activation patching and circuits. Causal mediation introduced the method [9]; ROME localized factual associations with causal tracing [1]; best practices are systematized in [10]. Head-level analyses include IOI [11] and copy suppression [12]. Closest to us, JuICE reports head superposition in knowledge conflicts [13]; we test its claim at head granularity in SLMs under a counterfactual-distractor regime and exploit the resulting taxonomy for intervention.

Inference-time intervention. ITI steers along probe-identified truthful directions [3]; truth directions are linearly represented [4]; hallucination can also be detected from sampling uncertainty [17]. Our interventions are *subtractive* (head knockout/mean-ablation) and selected by causal patching rather than probes, with explicit capability trade-off curves.

Attention sinks and knowledge editing. Autoregressive models allocate disproportionate attention to initial tokens ("attention sinks"), which stabilize generation but are usually treated as functionally inert no-ops [15]. Our findings add a twist: the conflict-specialist heads that dominate the CM-conflict circuit are BOS-sink heads that are causally *active* under misleading context (Sect. 4.2). Separately, knowledge editing locates and rewrites stored facts [1], [16]; we instead locate components involved in *context override* and ablate them at inference time, leaving parameters untouched.

3. Experimental Setup

Fig. 1 summarizes the protocol.

Models. Primary: Gemma-2-2B (26 layers, 8 heads). Cross-model: Llama-3.2-1B (16L/32H), Qwen2.5-1.5B (28L/12H). All experiments use bfloat16 on a single 24GB GPU; a parity check between our hooking framework and reference HuggingFace logits passed in float32 (mean KL 1.3×10^{-5}, argmax agreement 100%), bounding numerical noise two to three orders below measured effects.

Data. From CounterFact [1] we retain facts each model demonstrably knows (greedy completion equals the true target and the first-token logit gap over the counterfactual distractor ≥ 0.5), sampling 2,000 facts (16.7% survival of 12,000 scored) split 807/1,193 dev/held-out *by relation*, so the held-out split shares no relation with the split used for any selection. A CM-conflict suite asserts the distractor in three templates; a neutral control replaces it with a non-answer token.

Metrics and definitions. Let $z_t(x)$ and $z_d(x)$ denote the logits, at the answer position of input x, of the first tokens of the true target and the distractor, respectively. The *logit difference* is $\mathrm{LD}(x) = z_t(x) - z_d(x)$. Let c, m, and n denote the clean, misleading, and neutral prompts for a given fact, and let h index a specific attention head. Writing $x \leftarrow y_h$ for the input x with head h's activation replaced by its value from input y, the three directed patching effects are:

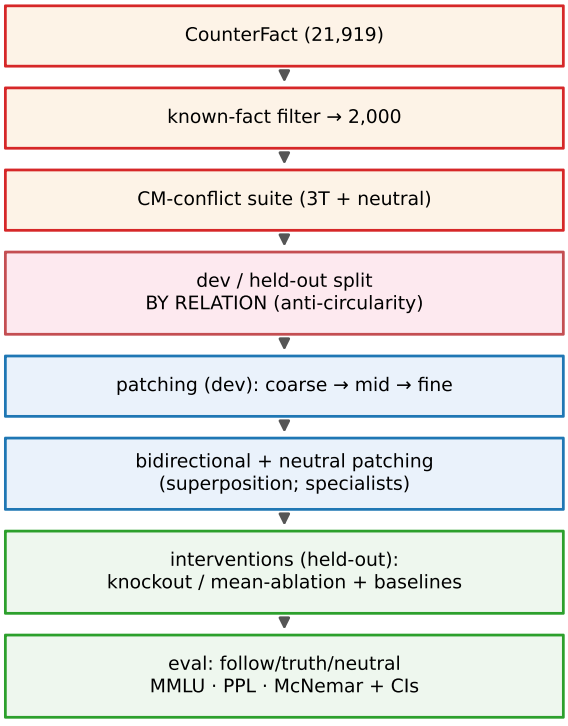


Fig. 1 Protocol overview. Data construction (top), causal localization on the dev split (middle), and interventions evaluated exclusively on held-out relations (bottom).

$$D = \mathrm{LD}(c) - \mathrm{LD}(m), \tag{1}$$

$$A_h = \frac{\mathrm{LD}(m \leftarrow c_h) - \mathrm{LD}(m)}{D}, \tag{2}$$

$$B_h = \frac{\mathrm{LD}(c) - \mathrm{LD}(c \leftarrow m_h)}{D}, \tag{3}$$

$$N_h = \frac{\mathrm{LD}(n \leftarrow c_h) - \mathrm{LD}(n)}{\mathrm{LD}(c) - \mathrm{LD}(n)}. \tag{4}$$

Here A_h is the *restore effect* (how much patching the clean value into the misleading run restores the true-answer margin; $A = 1$ is full restoration), B_h is the *context-push effect* (how much patching the misleading value into the clean run degrades the margin; $B = 1$ is full degradation), and N_h is the *neutral effect* (patch into the neutral-context run, normalized by the clean–neutral gap, *not* the clean–misleading gap D used for A and B). The *superposition index* $S = |A - B|/(|A| + |B| + \epsilon)$, with $\epsilon = 10^{-8}$, equals 1 for a head that pushes one side only and 0 for a fully dual-role head; when both $|A|$ and $|B|$ are near zero the index is uninformative, and we exclude such heads from superposition-based claims. Effects where the denominator has magnitude $< 10^{-6}$ are set to 0. Head-level values of A, B, and N are means over per-fact effects; the reported neutral ratio N/A is the ratio of these means (not a mean of per-fact ratios). Hallucination rate is the fraction of misleading-

prompt items whose greedy completion equals the distractor (context-following); accuracy is the fraction matching the true target. We use the term "misleading-context distractor-following" for this specific, programmatically verifiable metric, which is a strict subset of open-ended factual hallucination.

Protocol. Stage 1 (coarse): patch residual-stream block outputs layer-by-layer under two corruptions—ROME-style embedding noise, and the misleading context itself—measuring the normalized restoration of the true-vs-distractor logit difference. Stage 2 decomposes attention vs. MLP outputs on top layers; Stage 3 patches per-head z-vectors. Stage 4: bidirectional and neutral-context patching characterize each top head (A, B, N; S as above). Stage 5 intervenes by zeroing or mean-ablating selected heads and evaluates hallucination rate, accuracy, and neutral accuracy on held-out items; MMLU (2,000-question fixed subset) and WikiText-2 perplexity (1024-token windows) measure capability under the same hooks. Fairness baselines ablate randomly selected heads (5 seeds) and top-$|z|$-norm heads. Statistics: bootstrap 95% CIs (10,000 resamples over items); two-sided exact McNemar tests against the no-intervention condition.

Reproducibility details. Misleading templates (distractor d, prompt p): T0 = "*Earlier you learned that the correct answer is d. p*"; T1 = "*The document states: p d. Now answer correctly: p*"; T2 = "*Hint from the user: I am sure it is d. p*". Neutral control replaces d with "unknown". Models use HuggingFace checkpoints `google/gemma-2-2b`, `meta-llama/Llama-3.2-1B`, `Qwen/Qwen2.5-1.5B` (base, not instruct-tuned) loaded in bfloat16 via TransformerLens 2.x with `center_unembed=False`, `fold_ln=False`. Head ablation zeroes the head's z-vector at *all* token positions; mean-ablation replaces it with the mean z over 64 dev prompts. MMLU uses the `cais/mmlu` "all" test split, 0-shot, scored by comparing logits of the letters "A"–"D" at the answer position [20]; WikiText-2 uses the `wikitext-2-raw-v1` test split, 100 windows of 1024 tokens [21]. The parity check between TransformerLens and HuggingFace logits was conducted in float32 (mean KL 1.3×10^{-5}); all experiments run in bfloat16, whose measured noise floor (mean KL 1.2×10^{-3}) is 2–3 orders of magnitude below the smallest reported restoration effects (0.009–0.333).

4. Results

4.1 Behavioral landscape and localization

On known facts, the bare-prompt accuracy of Gemma-2-2B is 100%. Under misleading context the competition is nearly balanced overall (47.3% context-following vs. 46.2% truth; CI 46.0–48.5) but strongly template-dependent (24.2%/98.4%/19.2% following across templates), matching persuasion-strength variation reported behaviorally [7]; the neutral control stays at 87.4% truth with 1.2% copying of the neutral token, bounding pure token-copy effects (Table 1). Template 1, which places the distractor immediately before the repeated question, drives following to 98.4%—an upper bound on recency/copy contributions that the neutral control and bidirectional analyses below separate from belief override. Reference benchmarks (lm-eval-harness, full MMLU test set): MMLU 49.4%±0.4, TruthfulQA MC1/MC2 24.5%/36.3%, WikiText-2 PPL 65.3 (1024-token windows). The 2,000-question MMLU subset used for intervention evaluation yields 50.0% at baseline; the two scores differ because of subset sampling, not protocol error.

Table 1 Behavioral landscape on known facts (Gemma-2-2B, n=2,000; misleading suite per template).

condition	context-following	truth
clean (no context)	—	100%
misleading T0	24.2%	66.4%
misleading T1 (adjacent distractor)	98.4%	1.1%
misleading T2	19.2%	71.0%
neutral control	1.2%[†]	87.4%

[†]fraction copying the neutral token.

Table 2 Block-level decomposition (mean restoration effect, top layers).

block	noise regime	misleading regime
L22 attention	0.074	**0.121**
L25 attention	0.018	0.048
L21 MLP	**0.076**	≤0.01
L20 MLP	0.058	≤0.01

Fig. 2 shows the coarse maps. Restoration under the misleading regime concentrates in the final layers, peaking at the last layer (0.333); the noise regime peaks earlier (layer 21, 0.227). Table 2 decomposes the top blocks: the misleading-conflict signal is attention-dominated (L22 attention 0.121 vs. ≤0.01 for every MLP block), whereas pure recall restoration under noise mixes MLP and attention almost evenly. This contrast is itself informative: the classical ROME-style picture of factual recall as MLP-stored knowledge [1] does not transfer to the conflict regime, where what an intervention can restore flows through attention. Head-level patching then identifies a compact set: {L22.H0 (0.26), L25.H4 (0.24), L20.H3 (0.15), L25.H6 (0.12)} under noise, and {L25.H4 (0.08), L22.H4 (0.07), L22.H0, L20.H3} under the misleading context—note the partial overlap: three of four noise-top heads reappear in the conflict ranking, and L22.H4 emerges only under conflict.

4.2 Superposition and conflict specialists

Bidirectional patching (Fig. 4) shows every influential head conveys the misleading context more strongly than it restores the memory ($B > A$; superposition indices 0.21–0.62), confirming head-level superposition [13] in this regime. The neutral control separates two populations: L25.H4 retains 76% of its effect under the neutral prompt (generic memory role), whereas L22.H4, L22.H0 and L20.H3 show $N \approx 0$—

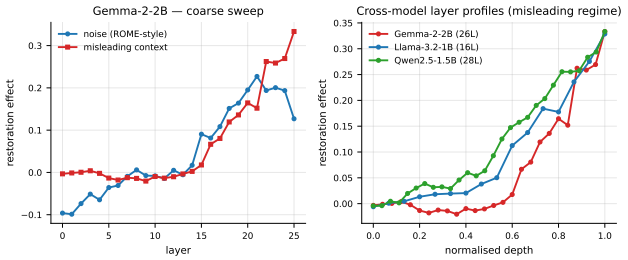


Fig. 2 Layer profiles of restoration effect. (a) Gemma-2-2B under both corruption regimes; (b) misleading-regime profiles across the three architectures at normalized depth. Late-layer concentration replicates (pairwise Spearman 0.81–1.00).

Table 3 Per-head characterization (Gemma-2-2B, top-10 by restore effect A; S = superposition index; N/A = neutral ratio).

head	A	B	N	S	N/A
L25.H4	0.081	0.134	0.062	0.24	0.76
L22.H4	0.071	0.143	−0.006	0.34	−0.09
L22.H0	0.048	0.095	0.002	0.33	0.05
L20.H3	0.038	0.096	−0.004	0.43	−0.10
L23.H7	0.016	0.038	−0.012	0.41	−0.72
L23.H4	0.012	0.022	−0.013	0.29	−1.02
L20.H7	0.010	0.042	−0.009	0.62	−0.91
L22.H3	0.010	0.030	−0.013	0.50	−1.33
L25.H2	0.010	0.024	0.004	0.42	0.38
L21.H1	0.009	0.019	0.001	0.34	0.06

conflict specialists whose neutral-control effect is minimal. Fig. 3 shows that, on the misleading prompt, the specialists concentrate their attention on the beginning-of-sequence token (0.92–1.00; we note that BOS precedes the misleading context in the causal decoder, so this pattern limits *what the head reads* rather than explaining *how* it conveys context information—the mechanism may operate through the value vectors at BOS, through the removal of attention from other positions, or through downstream residual-stream effects). The dual-role heads L25.H4/L25.H6, by contrast, attend to the template tokens asserting the distractor (0.67–0.78). We note that Fig. 3 shows a single representative example; the pattern is qualitatively stable but should be quantified across the full dataset in future work. Table 3 reports the full per-head characterization for the ten highest-restore heads. Two structure-level observations follow. First, $B > A$ for *every* head—no influential head is a pure "memory carrier"; all participate in conveying the context. Second, the specialist cluster (L22.H4, L22.H0, L20.H3) combines $N \approx 0$ with mid-range superposition indices (0.33–0.43), whereas strongly superposed heads with large $|N/A|$ (L20.H7, L22.H3) sit lower in the restore ranking: generic-role heads tend to be both superposed and less useful as intervention targets.

4.3 Interventions and quantified trade-offs

Table 4 and Fig. 5 report the held-out interventions. Knocking out the dev-ranked top-k heads reduces context-following monotonically (19.3%→1.6% at k=20) but at matching accuracy cost—a blunt instrument. The Stage-4 taxonomy, however, predicts that removing a conflict specialist should suppress the misleading pull without collateral damage. This is borne out: knocking out the single head L22.H4 reduces hallucination to 12.6% (McNemar b=81, c=1, $p < 10^{-23}$, two-sided exact McNemar b=81, c=1) while *increasing* accuracy to 79.7% (+5.3 points) and leaving neutral accuracy at 92.0% (baseline 93.5%). By contrast, ablating the five highest-norm heads *increases* hallucination to 33.0% (b=9, c=173): they are the dual-role workhorses, and their removal tips the final balance toward the context. Random-head ablation produces erratic, non-significant effects.

Table 4 Held-out interventions (Gemma-2-2B, misleading template T0). n=1,193 misleading + n=1,193 neutral items per config.

config	follow↓	truth↑	neutral
none	19.3%	74.4%	93.5%
top-1	15.8%	66.8%	82.6%
top-2	11.7%	69.9%	80.1%
top-5	8.5%	44.7%	50.9%
top-20	1.6%	19.4%	25.9%
spec1 (L22.H4 only)	**12.6%**	**79.7%**	**92.0%**
spec3	10.9%	63.3%	68.1%
mean-abl. top-5	8.8%	46.4%	53.8%
topnorm-5	33.0%	59.1%	85.9%
random-5 (mean)	18.9%	—	—

Capability costs are dissociated (Table 5): MMLU retention is statistically indistinguishable from baseline for every tested intervention (0.99–1.01), whereas perplexity rises 5–20%. The single-head specialist intervention is the cheapest of all: a 35% relative hallucination reduction—with an accuracy *gain*—at MMLU retention 0.998 and a perplexity cost of only 5%.

Template robustness. Re-evaluating the none, L22.H4, and top-5 conditions on the two remaining held-out templates tests whether the specialist's effect depends on how the context asserts the distractor. On T2 (user-hint template) L22.H4 again reduces following (16.6%→11.1%) while raising accuracy by 3.3 points, replicating T0. On T1—the extreme adjacent-distractor template whose baseline behavior is nearly pure context-following (98.4%) and which the neutral-control analysis attributes largely to recency copying—L22.H4 reduces following from 98.4% to 89.4% (9.0 percentage points; 9% relative) and raises accuracy from 0.8% to 8.7%, while top-5 reduces following from 98.4% to 58.3% (40.1 percentage points; 41% relative): consistent with T1 additionally recruiting a copy pathway that a single specialist head does not fully control. The intervention thus generalizes across persuasion formats, with residual copy-mediated following as the honest boundary of the single-head approach.

4.4 Cross-model replication

Replicating the pipeline on Llama-3.2-1B and Qwen2.5-1.5B (600 known facts each; 300/150 facts for coarse/fine sweeps) confirms the core structural and interventional find-

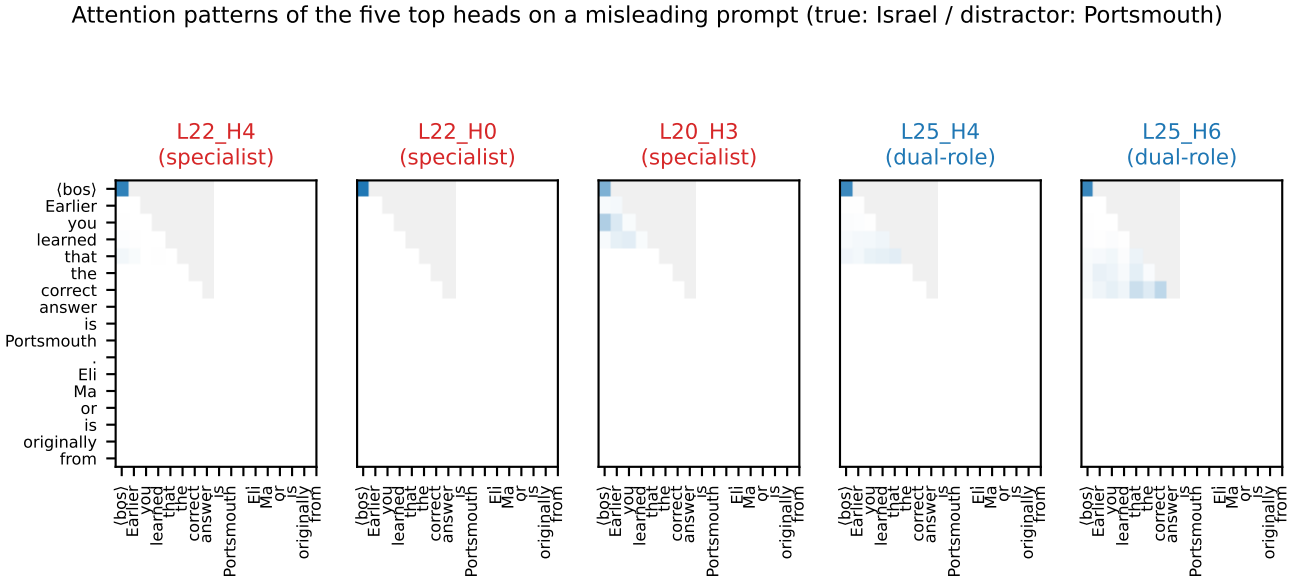


Fig. 3 Attention patterns of the five top heads on a representative misleading prompt ("Eli Maor is originally from"; true: Israel, distractor: Portsmouth). Conflict specialists (red, left three) attend almost exclusively to ⟨bos⟩; dual-role heads (blue, right two) spread attention over the template tokens asserting the distractor.

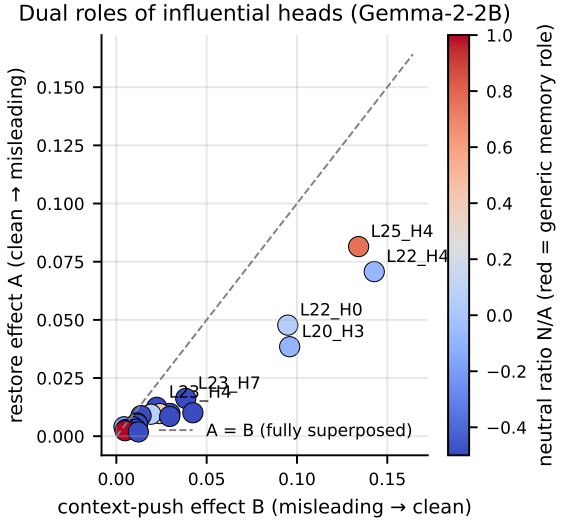


Fig. 4 Dual roles of the top-20 heads: context-push effect B vs. restore effect A, colored by neutral ratio N/A. Points near the diagonal are superposed; L22.H4 combines the strongest context conveyance with zero neutral role.

Table 5 Capability under intervention (fixed 2,000-question MMLU subset; PPL over 1024-token WikiText-2 windows).

config	MMLU	retention	PPL ratio
none	50.0%	1.00	1.00
top-2	49.7%	0.99	1.11
top-5	50.5%	1.01	1.20
spec1 (L22.H4 only)	**49.9%**	**1.00**	**1.05**
spec3	50.4%	1.01	1.16
mean-abl. top-5	50.0%	1.00	1.13

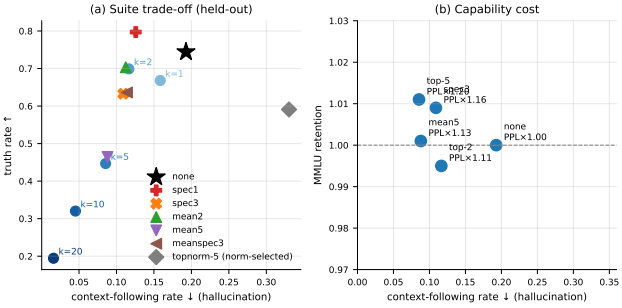


Fig. 5 Trade-offs. (a) suite space: hallucination vs. truth; the specialist point (red P) dominates the knockout frontier. (b) capability space: hallucination vs. MMLU retention (PPL ratios annotated).

ings, though the specialist taxonomy (Stage 4) was run only for Gemma-2-2B (Table 6). First, late-layer concentration holds in all three architectures: the most influential coarse layers are 20–25 of 26 (Gemma-2-2B), 13–15 of 16 (Llama), and 22–27 of 28 (Qwen), and the normalized-depth profiles rank-correlate at 0.81 (Gemma-2-2B vs. each other model) and 1.00 (Llama vs. Qwen). Second, baseline distractor-following rates vary across models (6.6% and 14.9%, both lower than Gemma-2-2B's 19.3% on the same held-out split), reflecting different context persuadability at different scales. Third, and most importantly, the single-head knockout from each model's own dev ranking reduces distractor-following by 30% (Llama) and 50% (Qwen) with accuracy changes of +0.5/+0.5 points and MMLU retention 1.01/1.00—the same qualitative signature as Gemma-2-2B's specialist result, obtained without any per-model specialist analysis, suggesting

Table 6 Cross-model replication summary (misleading regime, held-out).

	Gemma-2-2B	Llama	Qwen
layers	26	16	28
top coarse layers	20–25	13–15	22–27
facts used	2,000	600	600
follow (none)	19.3%	6.6%	14.9%
follow (top-1)	12.6%*	4.6%	7.5%
Δ truth	+5.3*	+0.5	+0.5
MMLU retention	.99–1.01	1.01	1.00

*Gemma-2-2B uses the Stage-4-selected specialist (L22.H4); the other models use their own dev-ranking top-1. Gemma-2-2B retention spans the tested interventions (top-2/5, spec3, mean-abl.).

the effect is a property of the patching-guided selection rather than of one architecture. Per-head restoration effects are smaller in the 1–1.5B models (0.04–0.06 vs. 0.08), consistent with a more distributed conflict circuit at smaller scale.

5. Discussion

Mechanism. The CM conflict is resolved late and attentionally: answer-position representations in the final layers carry most of what an intervention can restore, and the influential heads are conduits of the misleading context at least as much as carriers of memory ($B > A$ throughout). "Restoration" by patching therefore partly works by displacing a head's context conveyance rather than injecting memory—the neutral-control analysis makes this distinction measurable, addressing copy-vs-recall confounds.

Attention-sink specialists. The most surprising finding is the identity of the conflict specialists: heads whose attention is almost entirely absorbed by the beginning-of-sequence token. In the streaming-efficiency literature such sinks are treated as stabilizing but semantically inert no-ops [15]. Our results show that a sink head can be causally significant in the CM-conflict regime: L22.H4 conveys the strongest misleading-context push (B=0.143) and its removal improves factual accuracy while leaving MMLU and perplexity nearly unchanged (retention 0.998, PPL ratio 1.05). We note, however, that $N \approx 0$ under a specific neutral-patching control does not prove the head has no other function; it shows only that this particular intervention produced minimal effect in the tested condition. The mechanism by which a BOS-attending head influences the conflict remains an open question—possible routes include value vectors at the BOS position, attention-weight redistribution away from informative tokens, or downstream residual-stream effects—and our single-example attention visualizations should be quantified across the dataset before drawing mechanistic conclusions. What the results do establish is that sink-focused heads should not be excluded from behavioral circuit searches.

Design implication. The dissociation between MMLU (unchanged within measurement precision on a 2,000-question subset) and perplexity (degraded 5–20%) suggests these heads implement contextual integration that matters for language modeling loss but not for knowledge-task accuracy on the tested benchmarks. The contrast with norm-based selection carries a practical warning: cheap proxies for "importance" can select exactly the wrong components and *worsen* the targeted behavior.

Limitations and future work. (i) English-only, single-decoding (greedy), completion-style evaluation; (ii) hallucination is operationalized as verifiable counterfactual override, not open-ended factuality; (iii) head selection is causal but the circuit description is coarse—path-level analysis (direct vs. mediated effects) is the natural next step; (iv) Gemma-2-2B is not the newest generation—chosen for tooling maturity and comparability with the interpretability literature; newer families (Gemma 3/4) and instruction-tuned variants are future work; (v) capability is proxied by MMLU/PPL subsets; (vi) the conflict-specialist analysis (Stage 4) was run only for Gemma-2-2B; extending the specialist taxonomy to the replication models, and testing whether entity popularity modulates circuit reliance, remain open.

6. Conclusion

We mapped and exploited the CM-conflict circuit of small language models: late-layer attention heads with measurable superposition, among which specialist heads (identified via bidirectional patching and a neutral-control analysis in Gemma-2-2B) can be knocked out individually—distractor-following falls by 30–50% relative, accuracy rises, and MMLU retention remains near 1.00 on the tested subset—while norm-based selection backfires. All selection was performed on dev relations with results reported on held-out relations, and the core findings (late-layer concentration, single-head intervention effectiveness) replicated across three architectures. Code, per-item records, and scripts are available from the author upon request; a public release is planned upon publication.

Acknowledgment

This work used cloud computing resources funded by the author's institutional research grant.

References

[1] K. Meng, D. Bau, A. Andonian, and Y. Belinkov, "Locating and editing factual associations in GPT," in *Proc. NeurIPS*, 2022.
[2] L. Yu, M. Cao, J. C. K. Cheung, and Y. Dong, "Mechanistic understanding and mitigation of language model non-factual hallucinations," in *Findings of EMNLP*, 2024.
[3] K. Li, O. Patel, F. Viégas, H. Pfister, and M. Wattenberg, "Inference-time intervention: Eliciting truthful answers from a language model," in *Proc. NeurIPS*, 2023.
[4] S. Marks and M. Tegmark, "The geometry of truth: Emergent linear structure in large language model representations of true/false datasets," in *Proc. COLM*, 2024.
[5] R. Xu, Z. Qi, Z. Guo, C. Wang, H. Wang, Y. Zhang, and W. Xu, "Knowledge conflicts for LLMs: A survey," in *Proc. EMNLP*, 2024.
[6] Z. Ji *et al.*, "Survey of hallucination in natural language generation,"

ACM Comput. Surv., 2023.

[7] K. Du, V. Snæbjarnarson, N. Stoehr, J. C. White, A. Schein, and R. Cotterell, "Context versus prior knowledge in language models," in *Proc. ACL*, 2024.

[8] Y.-S. Chuang, L. Qiu, C.-Y. Hsieh, R. Krishna, Y. Kim, and J. Glass, "Lookback lens: Detecting and mitigating contextual hallucinations in large language models using only attention maps," in *Proc. EMNLP*, 2024.

[9] J. Vig, S. Gehrmann, Y. Belinkov, S. Qian, D. Nevo, Y. Singer, and S. Shieber, "Investigating gender bias in language models using causal mediation analysis," in *Proc. NeurIPS*, 2020.

[10] F. Zhang and N. Nanda, "Towards best practices of activation patching in language models: Metrics and methods," in *Proc. ICLR*, 2024.

[11] K. Wang, A. Variengien, A. Conmy, B. Shlegeris, and J. Steinhardt, "Interpretability in the wild: A circuit for indirect object identification in GPT-2 small," in *Proc. ICLR*, 2023.

[12] C. McDougall, A. Conmy, C. Rushing, T. McGrath, and N. Nanda, "Copy suppression: Comprehensively understanding a motif in language model attention heads," in *Proc. BlackboxNLP*, 2024.

[13] G. Li, Y. Chen, and H. Tong, "Taming knowledge conflicts in language models," in *Proc. ICML*, 2025.

[14] J. Jiang, Z. Wu, Z. Wu, and X. Dai, "Causal evidence for attention head imbalance in modality conflict hallucination," arXiv:2605.19250, 2026.

[15] G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis, "Efficient streaming language models with attention sinks," in *Proc. ICLR*, 2024.

[16] Y. Yao, P. Wang, B. Tian, S. Cheng, Z. Li, S. Deng, H. Chen, and N. Zhang, "Editing large language models: Problems, methods, and opportunities," in *Proc. EMNLP*, 2023.

[17] S. Farquhar, J. Kossen, L. Kuhn, and Y. Gal, "Detecting hallucinations in large language models using semantic entropy," *Nature*, vol. 630, pp. 625–630, 2024, DOI: 10.1038/s41586-024-07421-0.

[18] M. V. Pham, H. Borkakoty, and Y. Hou, "Where knowledge collides: A mechanistic study of intra-memory knowledge conflict in language models," arXiv:2601.09445, 2026.

[19] D. Campregher, Y. Chen, S. Hoffman, and M. Heuss, "Tracing facts or just copies? Attention heads and counterfactual tasks," *Trans. Mach. Learn. Res.*, 2025.

[20] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt, "Measuring massive multitask language understanding," in *Proc. ICLR*, 2021.

[21] S. Merity, C. Xiong, J. Bradbury, and R. Socher, "Pointer sentinel mixture models," in *Proc. ICLR*, 2017.

[22] Gemma Team, "Gemma 2: Improving open language models at a practical size," 2024.

[23] A. Grattafiori *et al.*, "The Llama 3 herd of models," 2024.

[24] Qwen Team, "Qwen2.5 technical report," 2024.

[25] S. Lin, J. Hilton, and O. Evans, "TruthfulQA: Measuring how models mimic human falsehoods," in *Proc. ACL*, 2022.

[26] M. Zhang, O. Press, W. Merrill, A. Liu, and N. A. Smith, "How language model hallucinations can snowball," in *Proc. ICML*, 2024.